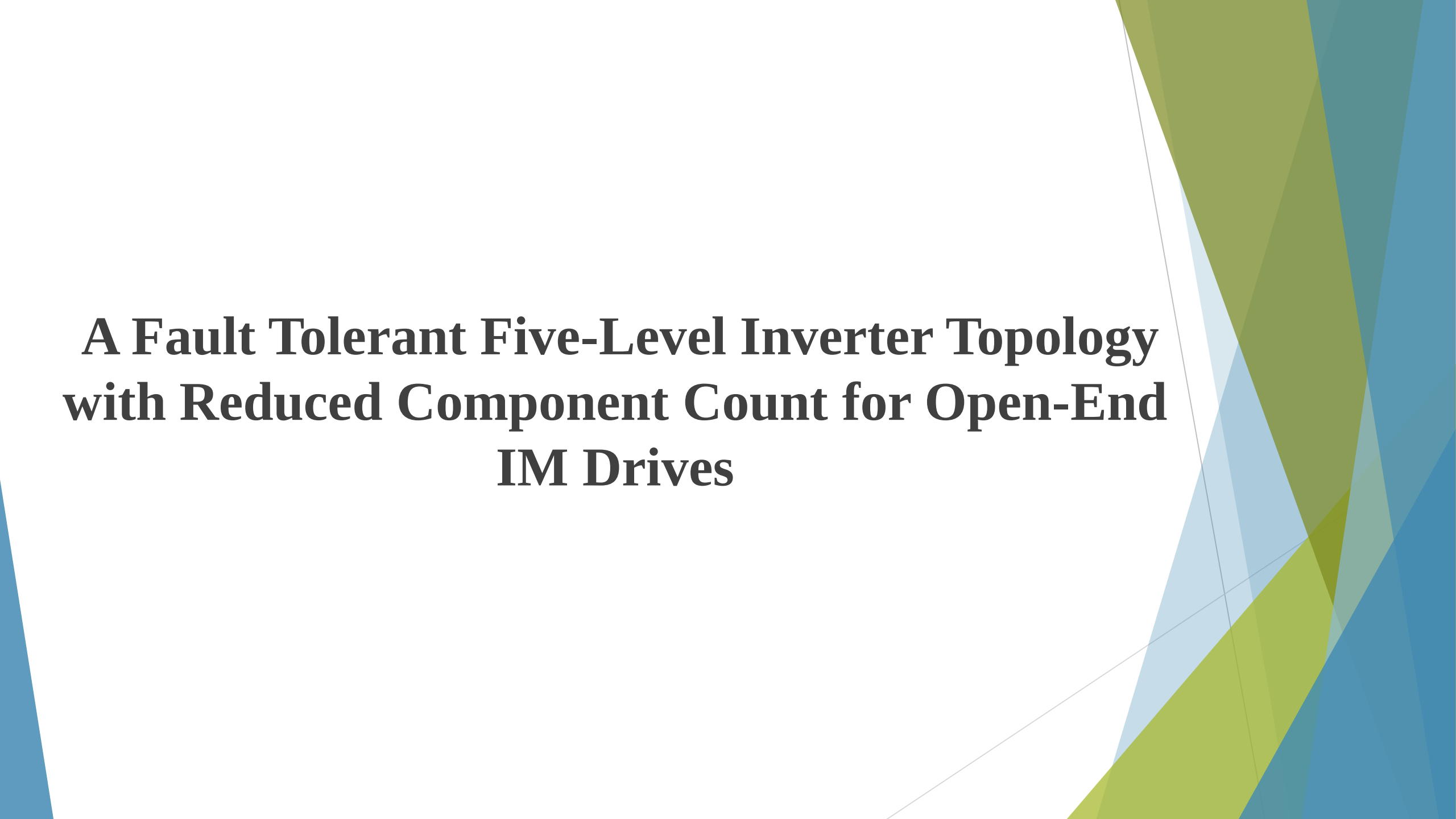




A Fault Tolerant Five-Level Inverter Topology with Reduced Component Count for Open-End IM Drives

EXTENSION SYSTEM:

- ❑ In the proposed system, we have utilized open loop multilevel is used. So, we are going to utilize closed loop FLC-Based Space Vector Pulse Width Modulation as extension to this project. So, that we can control the faults and the components count.

BLOCK DIAGRAM

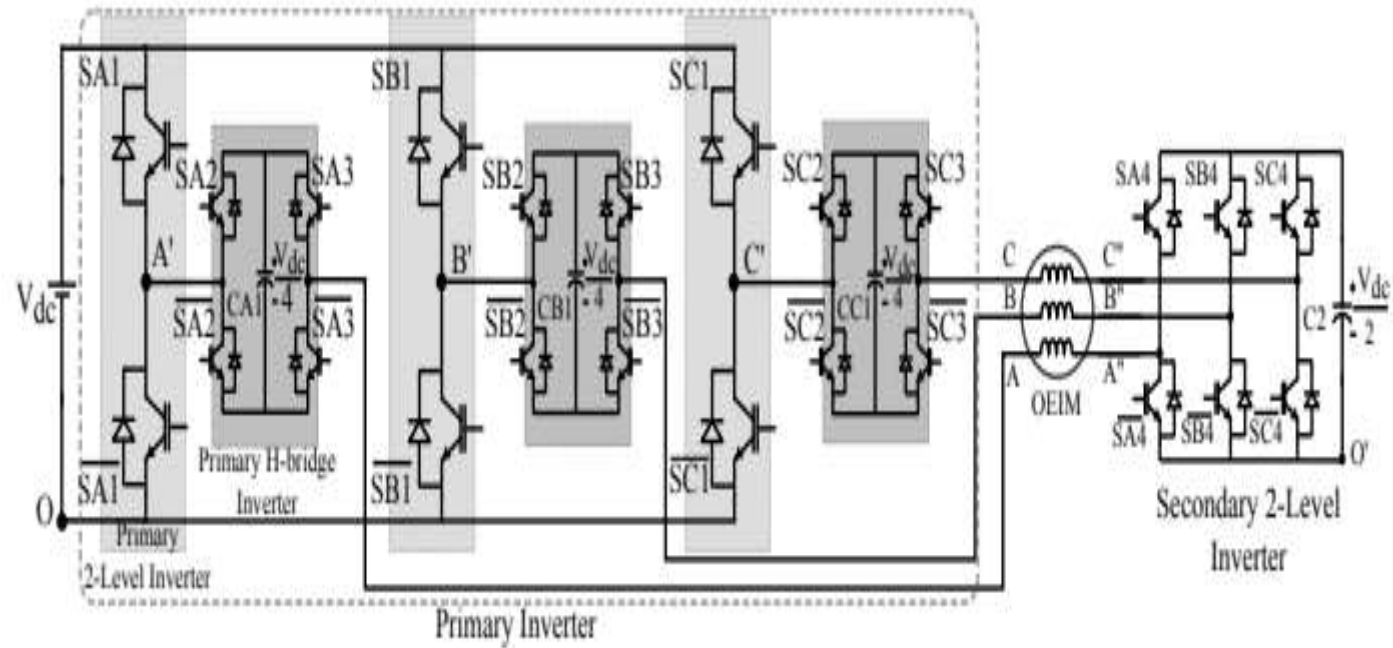


Fig. Block diagram of the proposed system

Software & Hardware Requirements

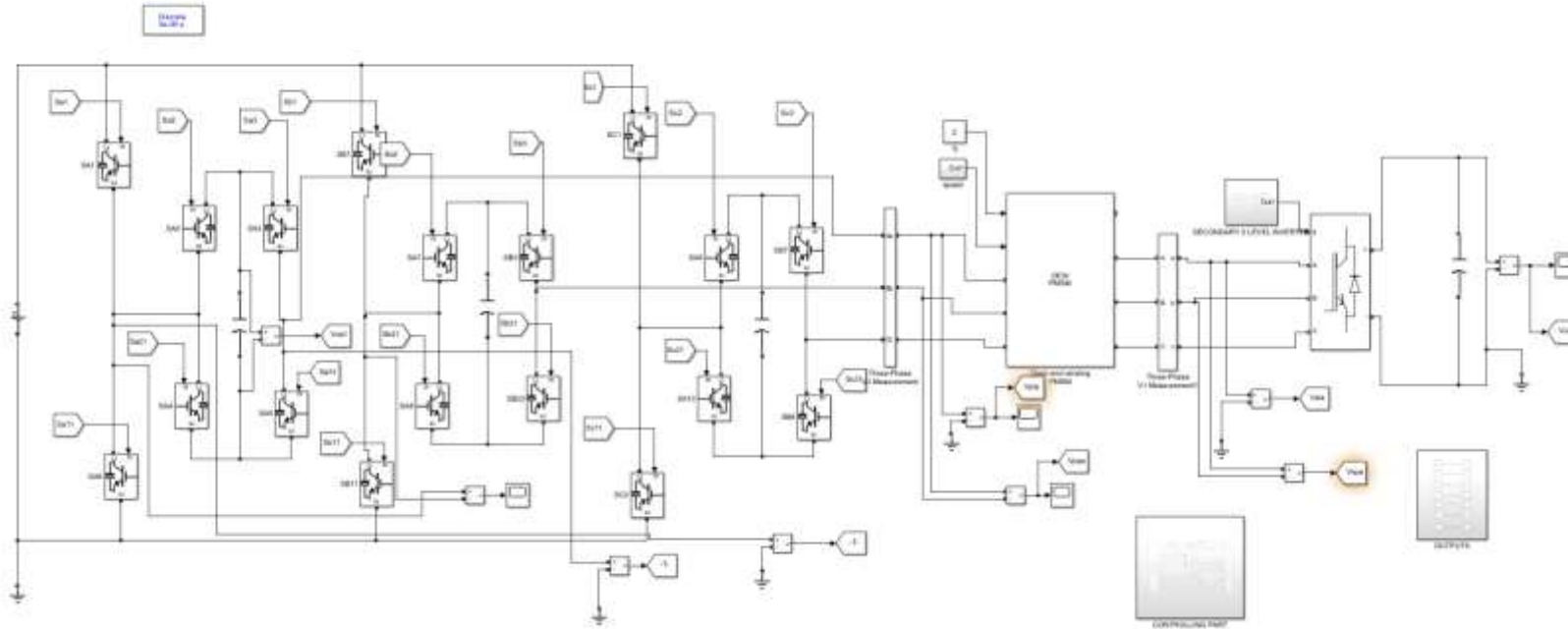
❑ Software configuration:

- ▶ Operating system : windows 7/8/10
- ▶ Application software: matlab/simulink

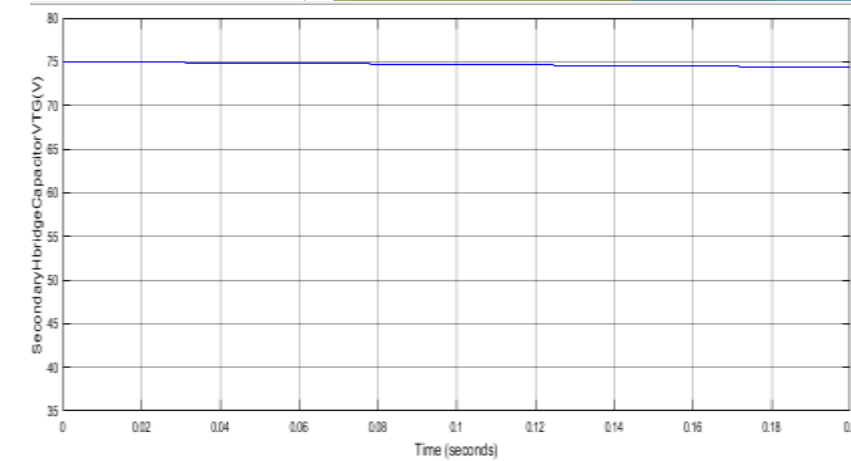
❑ Hardware configuration:

- ▶ Ram : 8 gb
- ▶ Processor : i3 / i5 (mostly prefer)

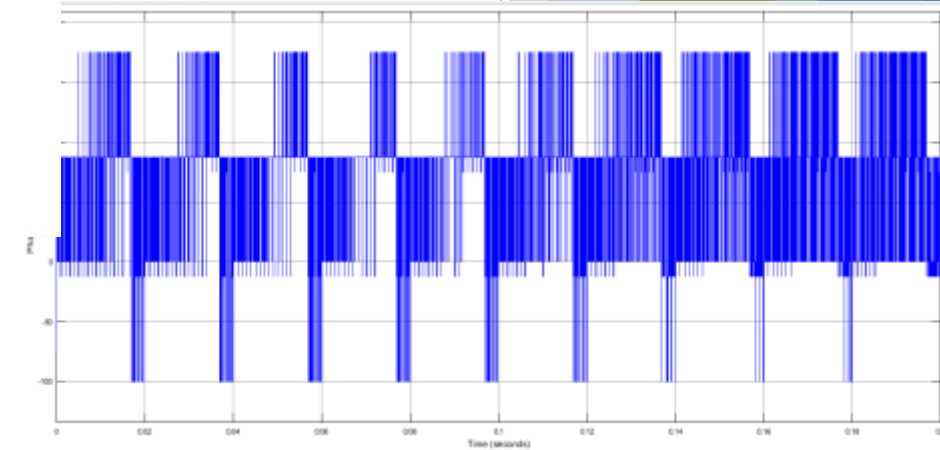
SIMULATION RESULTS



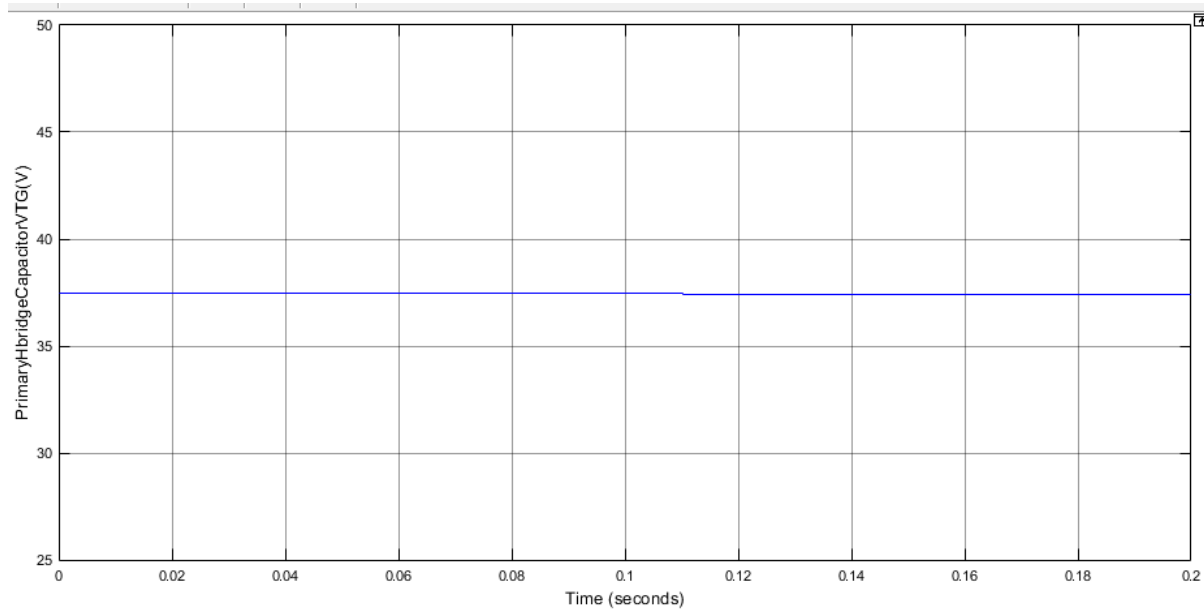
Schematic diagram of proposed 5-Level inverter topology with OEIM.,



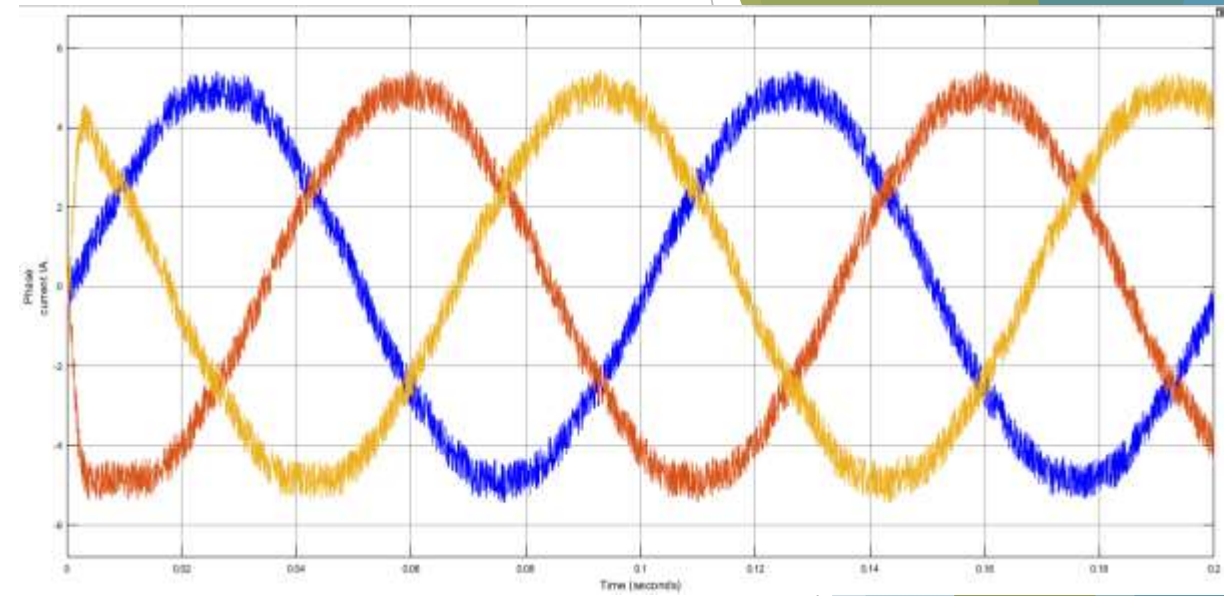
Secondary inverter capacitor voltage $VC2$,



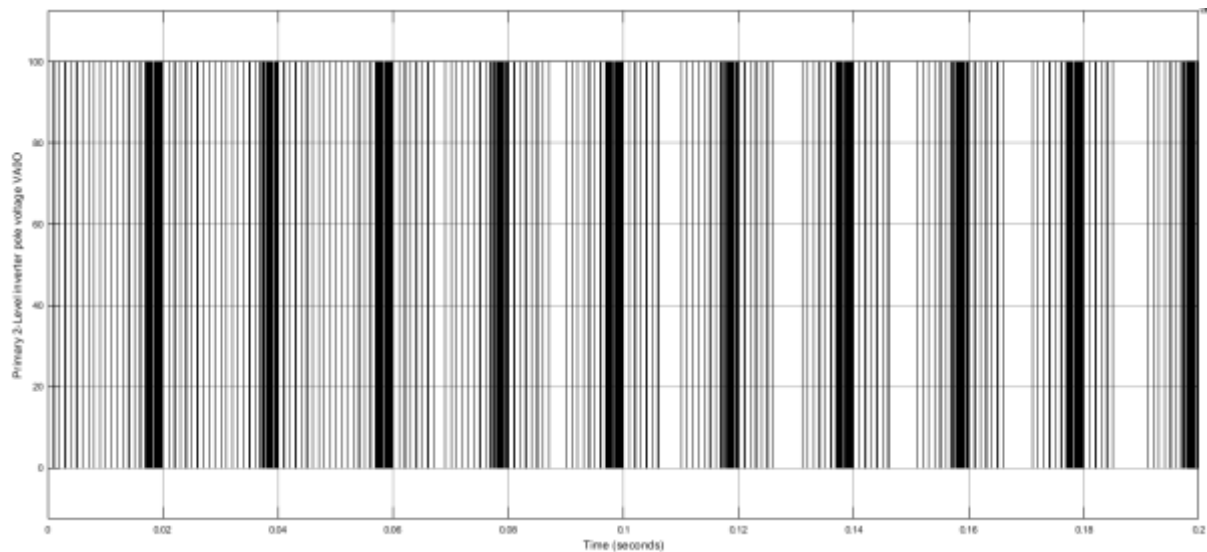
Phase Voltage VAA



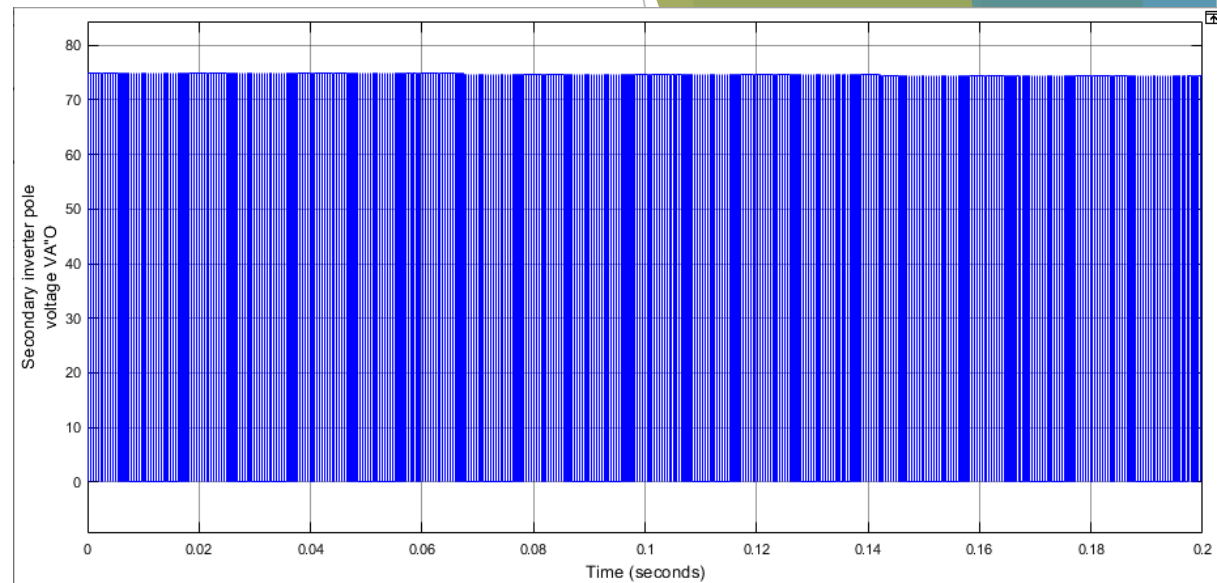
Primary H-Bridge inverter capacitor voltage VCA1



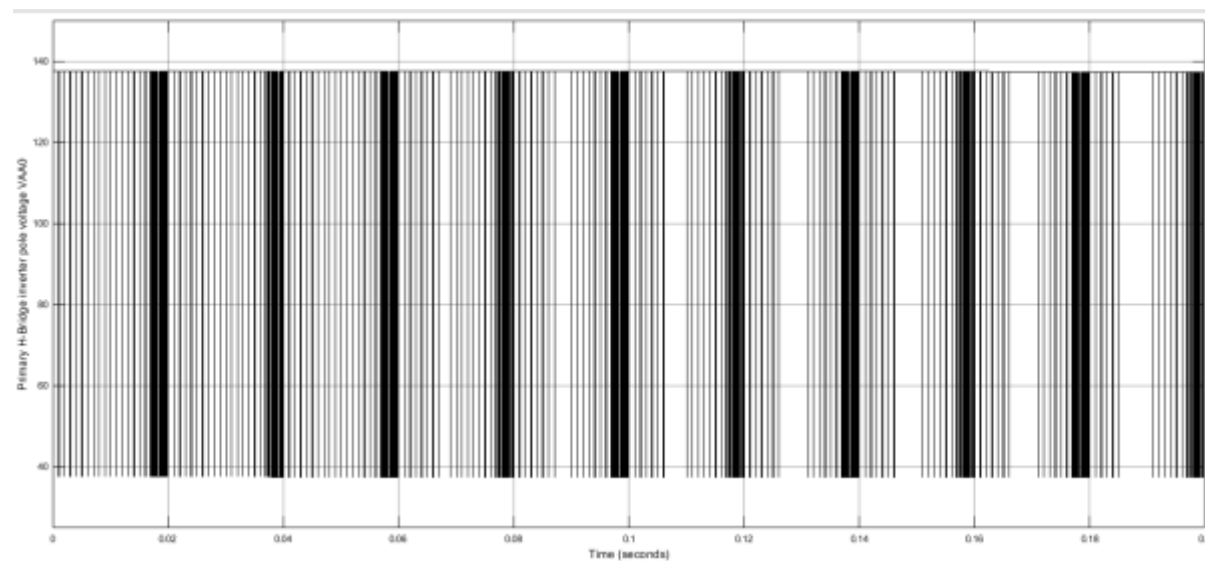
Phase currents(Iabc)



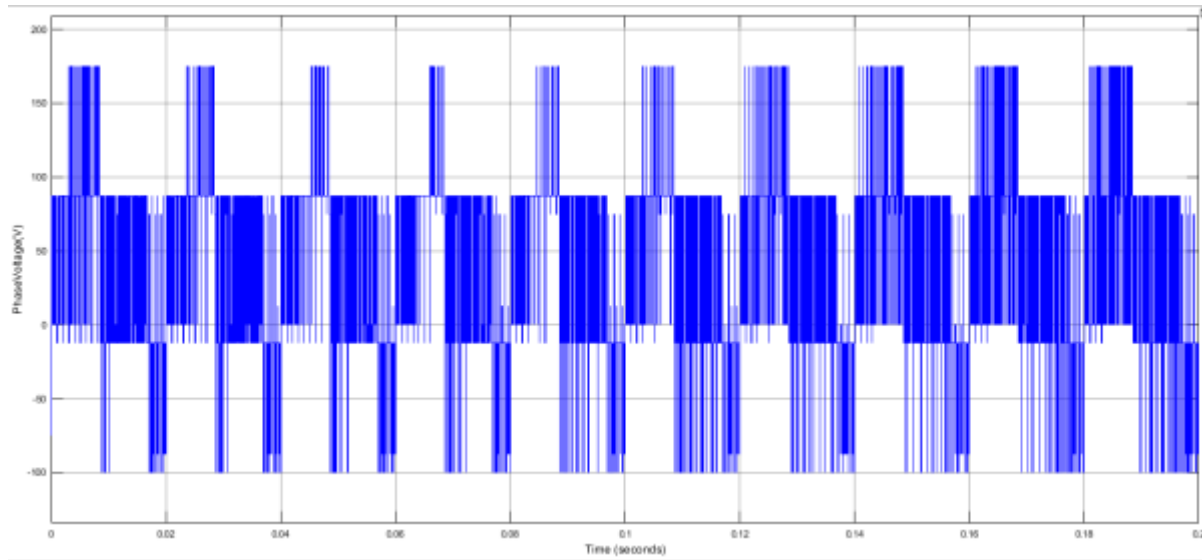
Primary 2-Level inverter pole voltage VA0



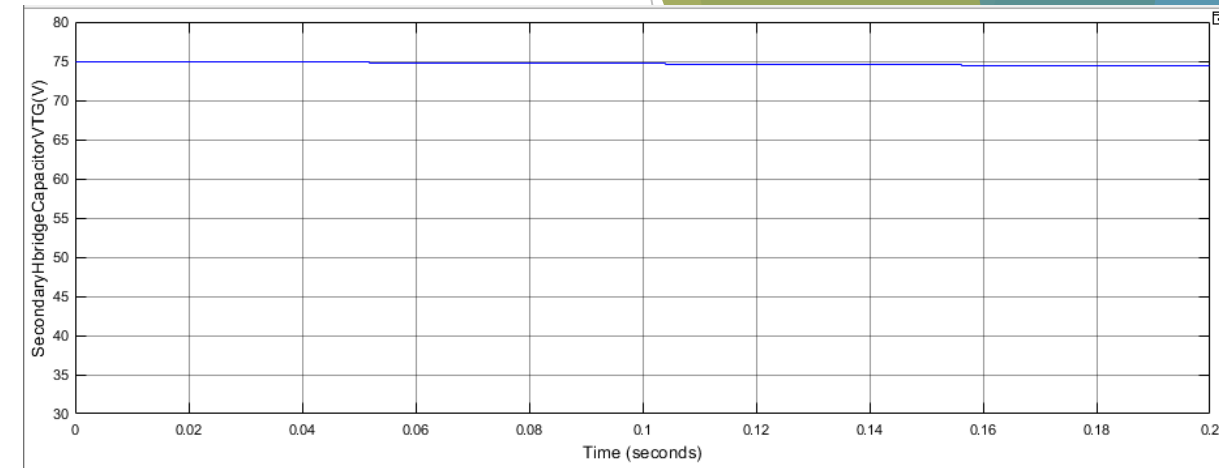
Secondary inverter pole voltage



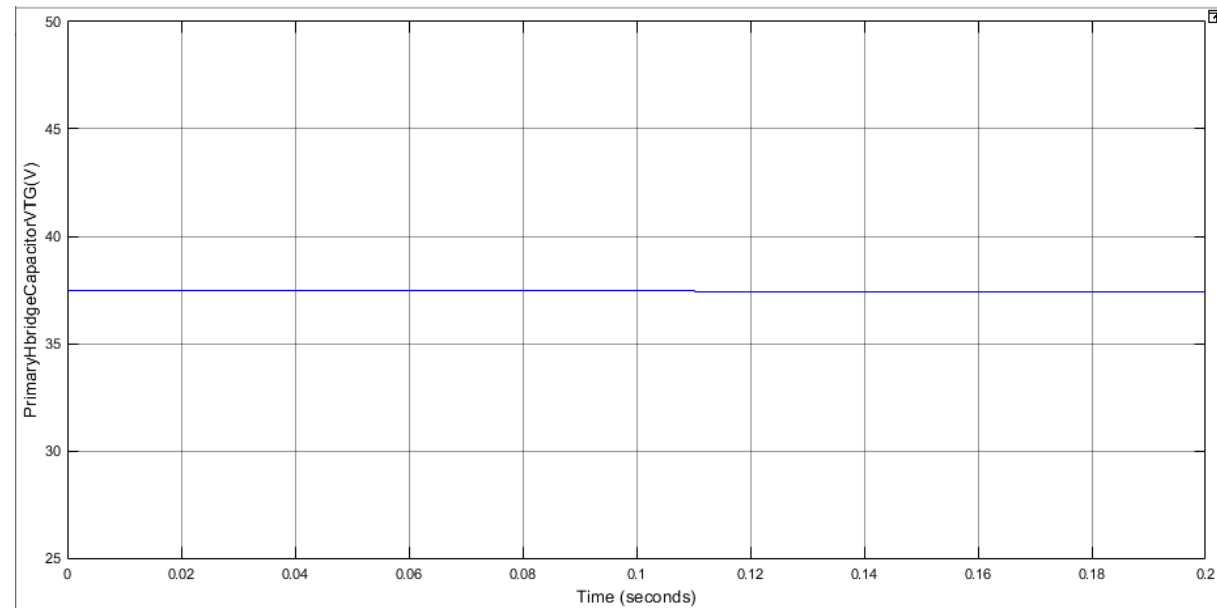
Primary H-Bridge inverter pole voltage VAA0



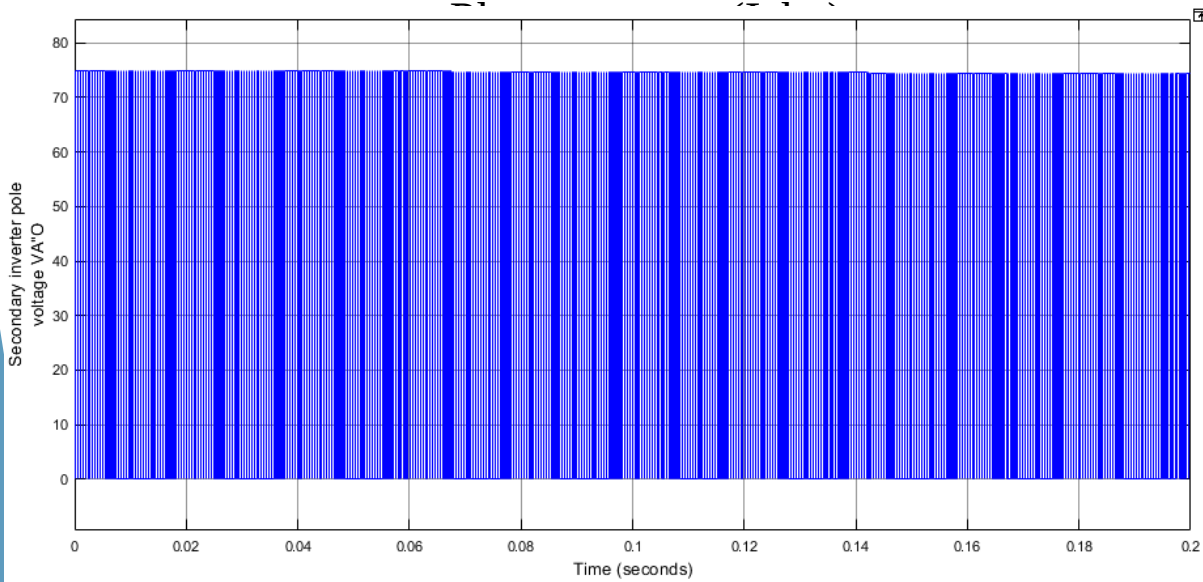
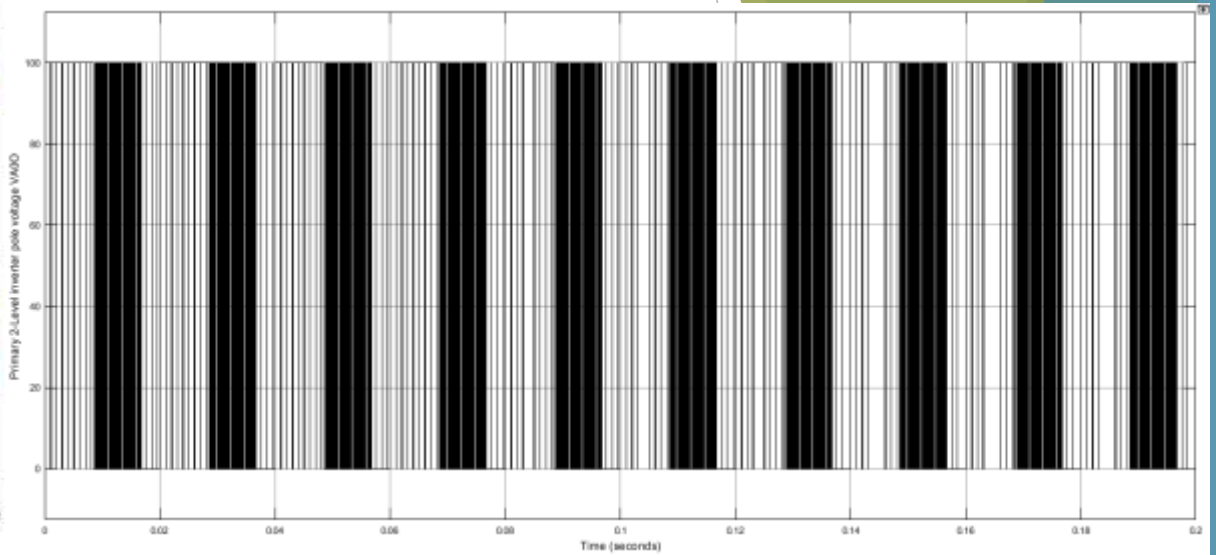
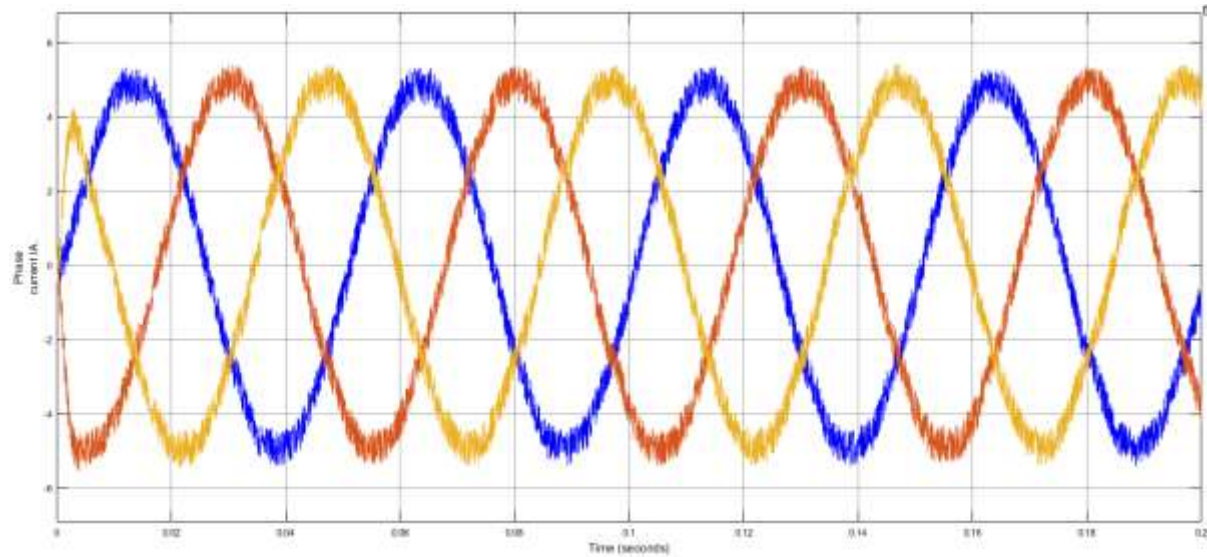
Phase Voltage VAA"



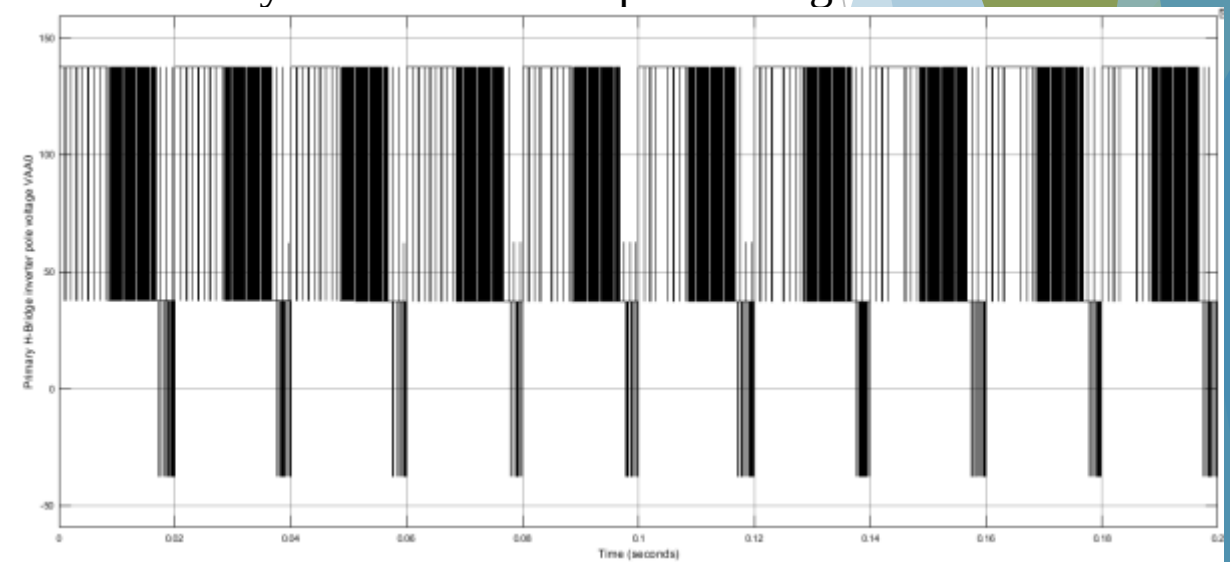
Secondary inverter capacitor voltage VC2,



Primary H-Bridge inverter Capacitor voltage VCA1



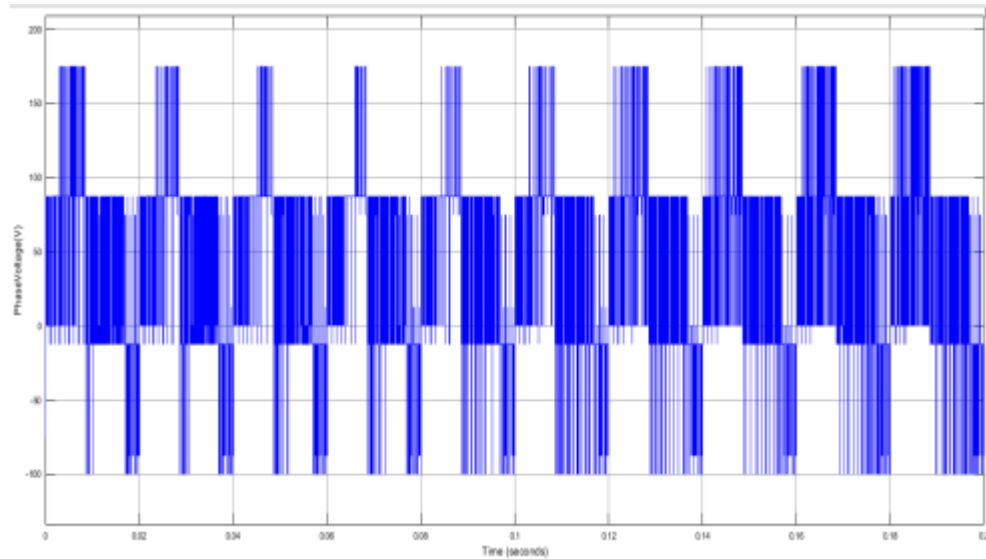
Primary 2-Level inverter pole voltage VA0



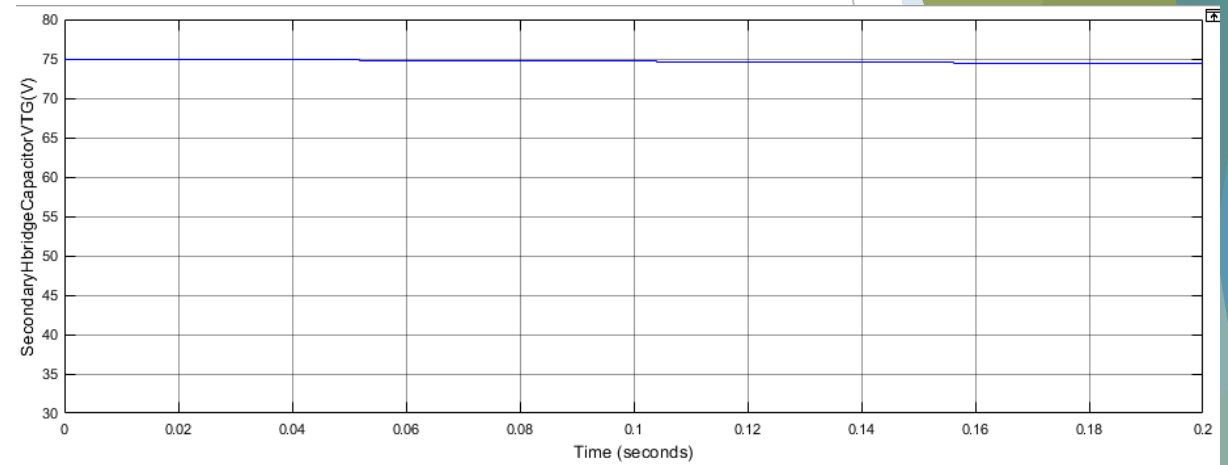
Primary H-Bridge inverter pole voltage VAA0

Secondary inverter pole voltage

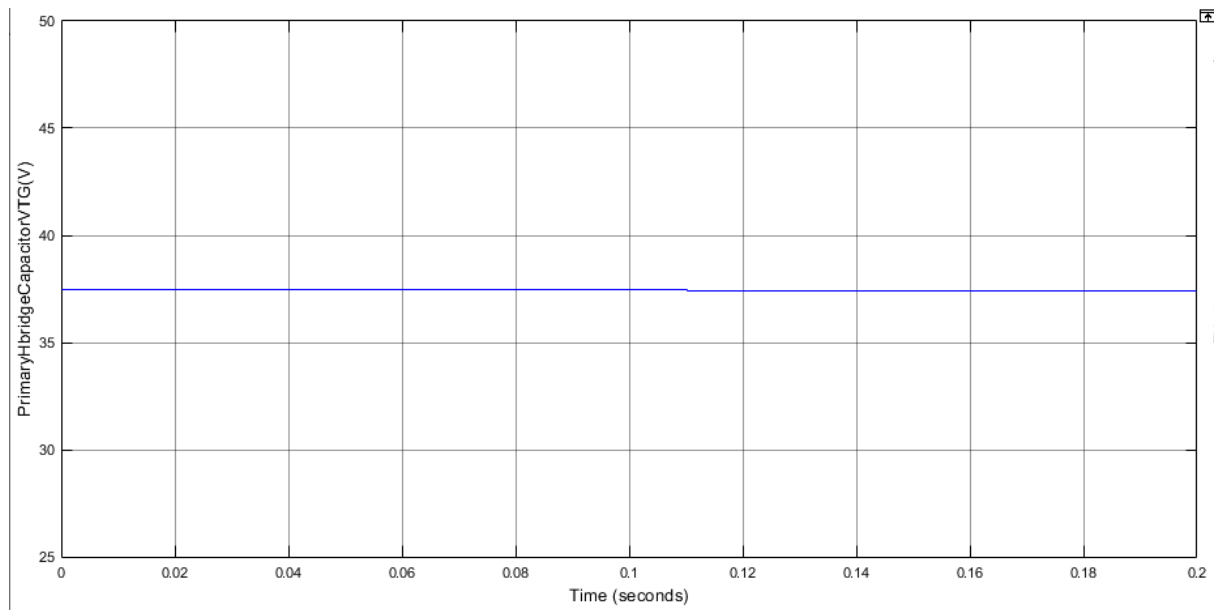
Case_3 at 45 Hz



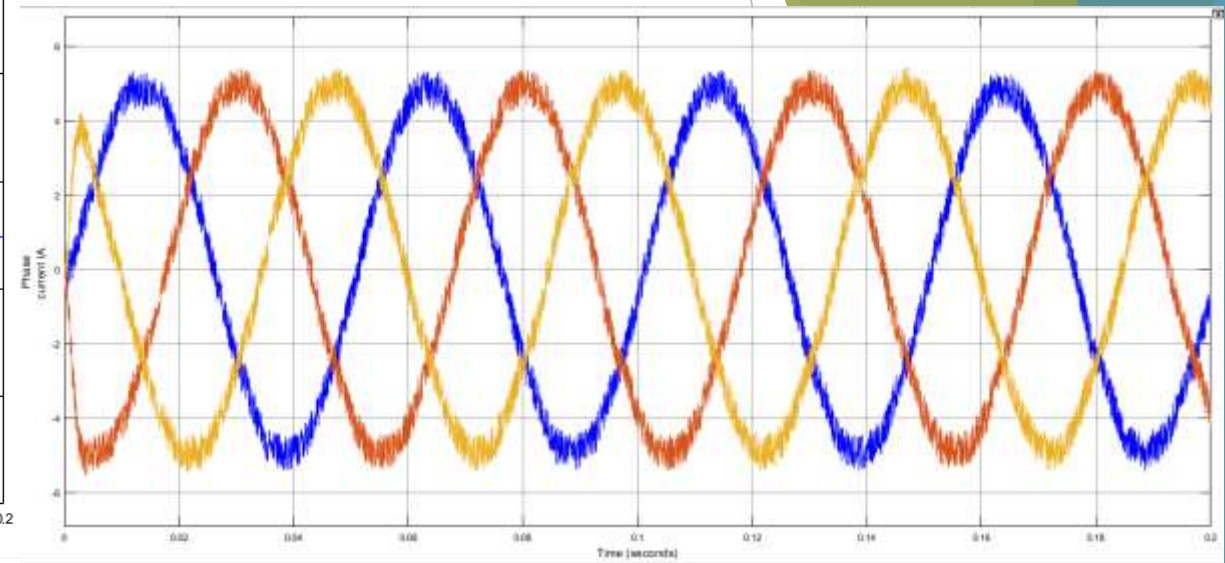
Phase Voltage VAA"



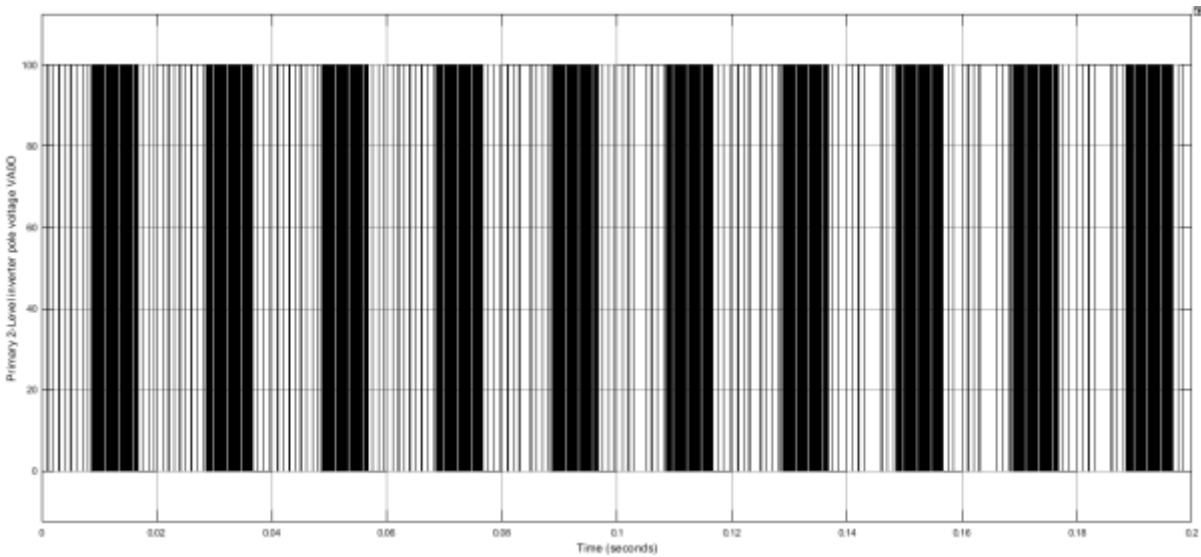
Secondary inverter capacitor voltage VC2,



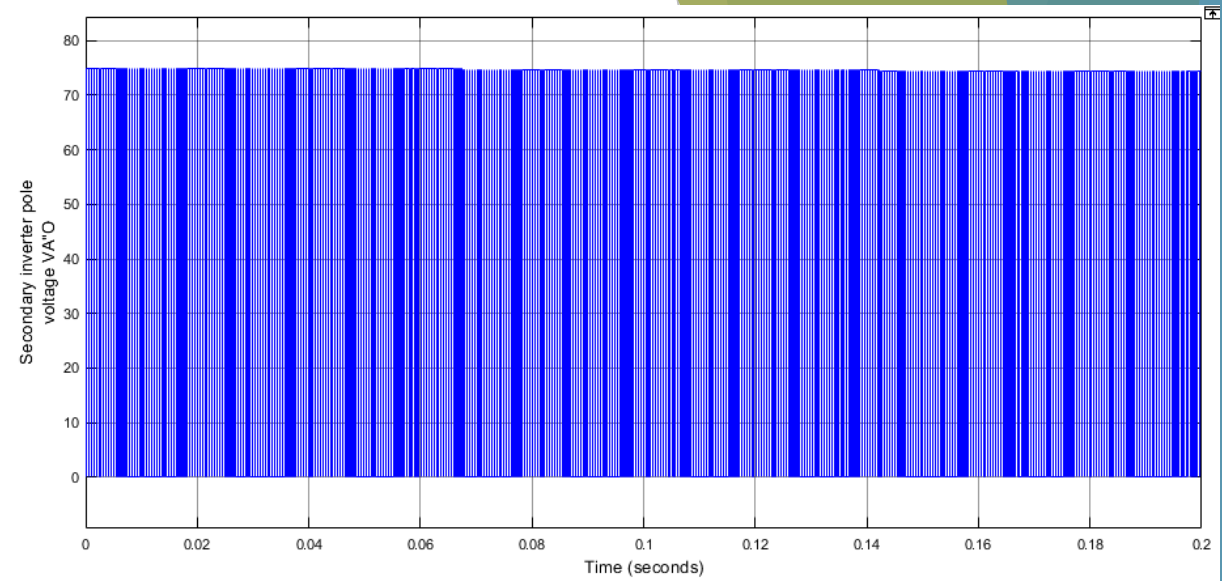
Primary H-Bridge inverter capacitor voltage VCA1



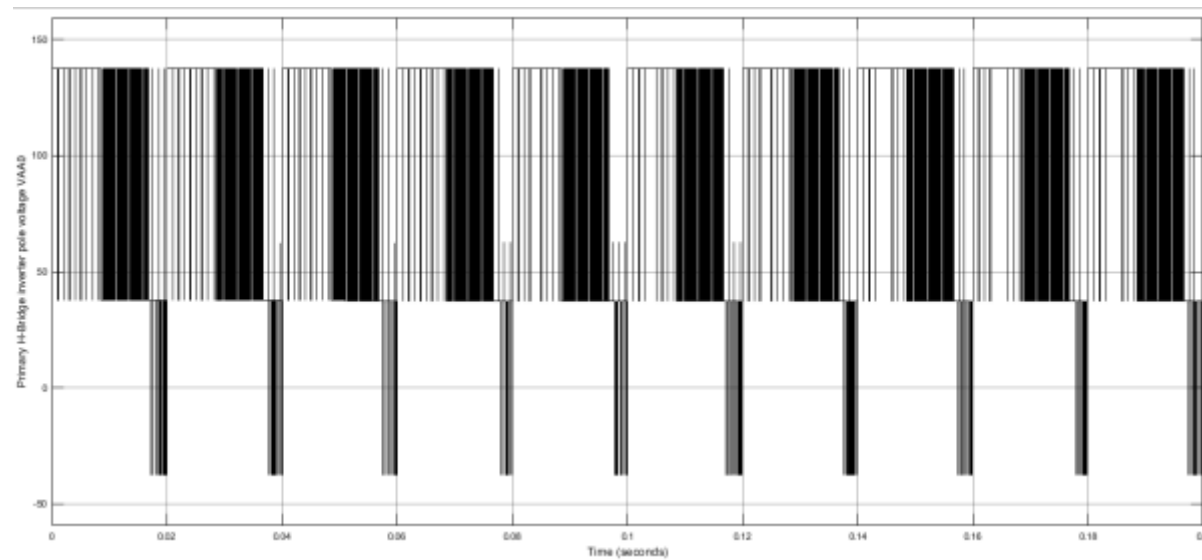
Phase currents(Iabc)



Primary 2-Level inverter pole voltage VA0

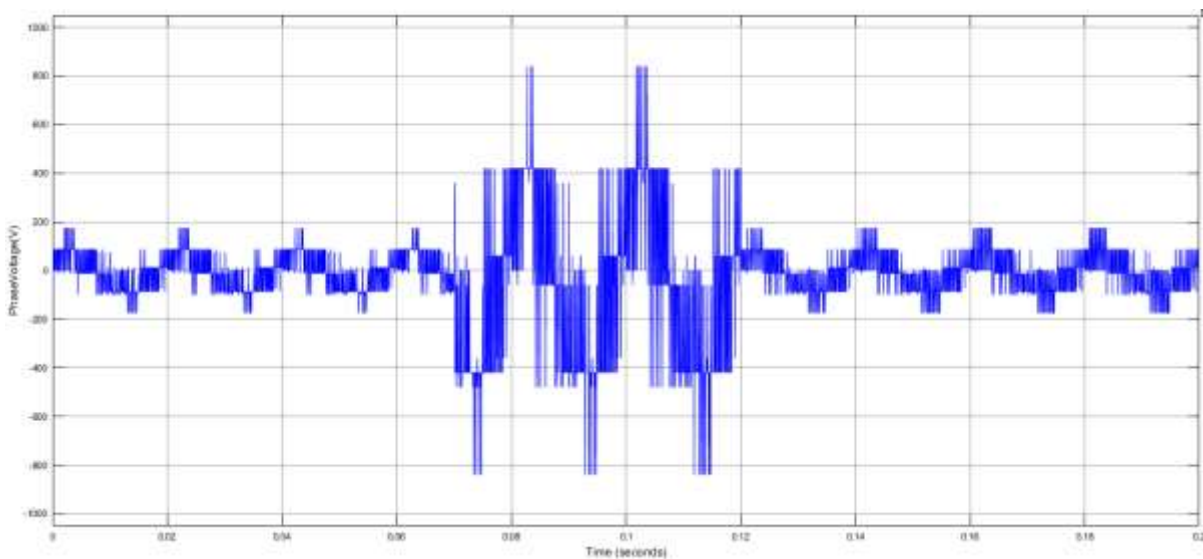


Secondary inverter pole voltage

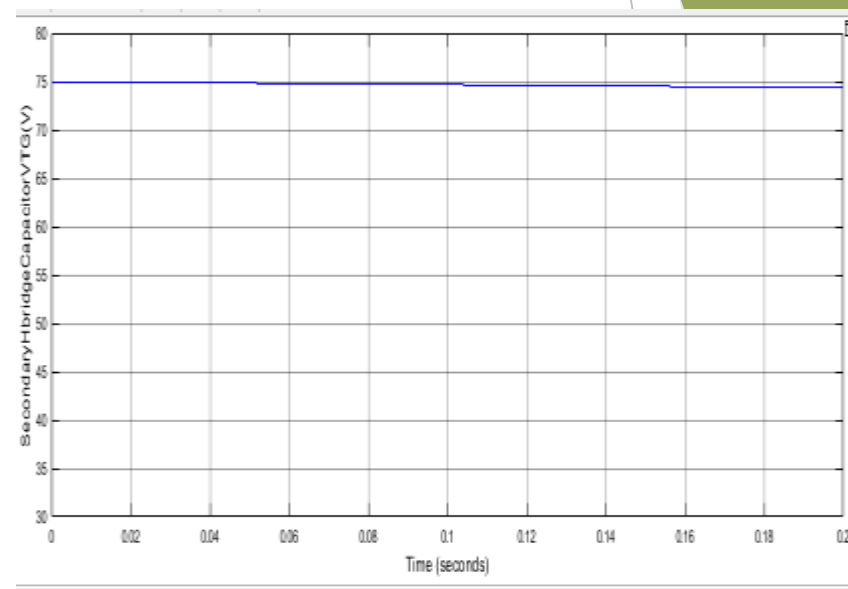


Primary H-Bridge inverter pole voltage VAA0

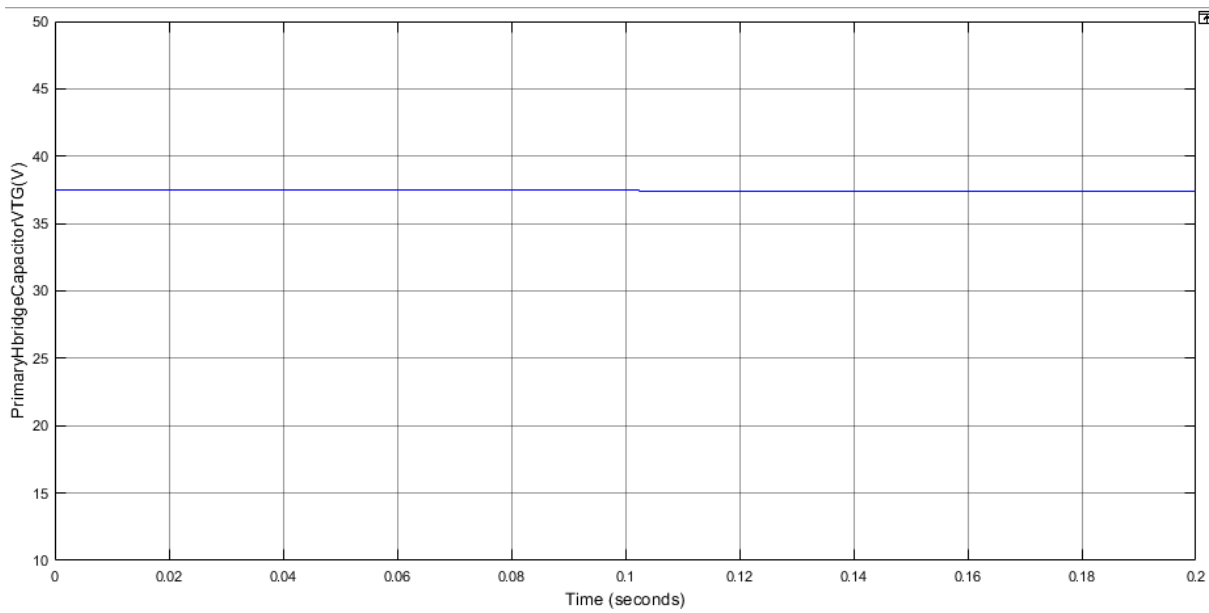
Demonstration of fault tolerance: (a) Fault in H-bridge after tf1 instant



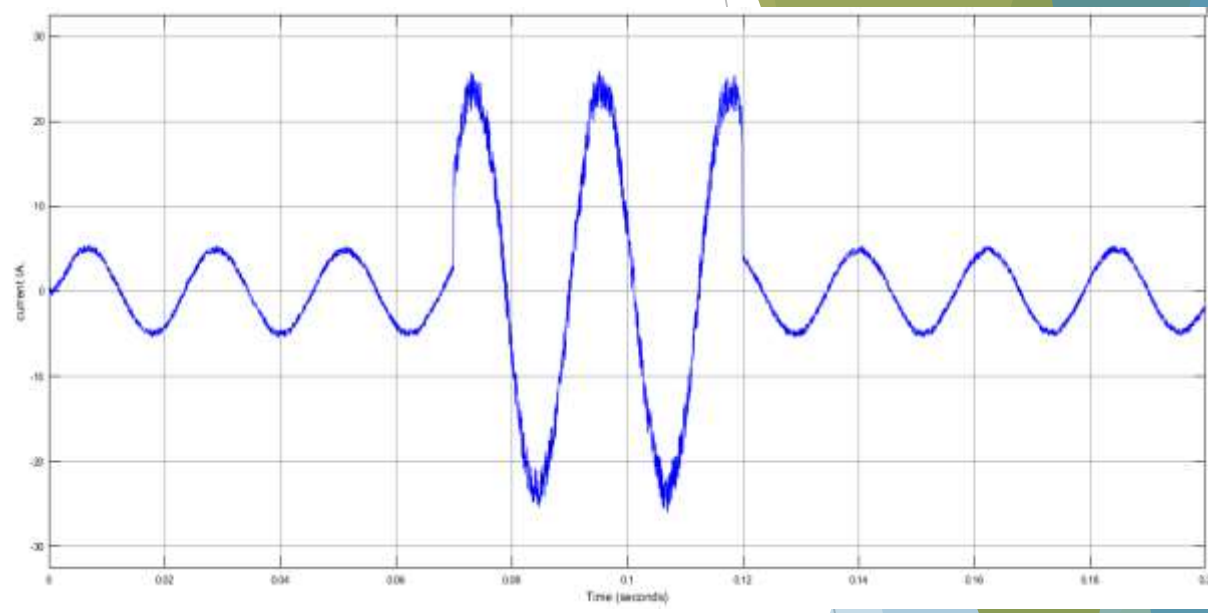
Phase Voltage



Secondary H Bridge Capacitor Voltage

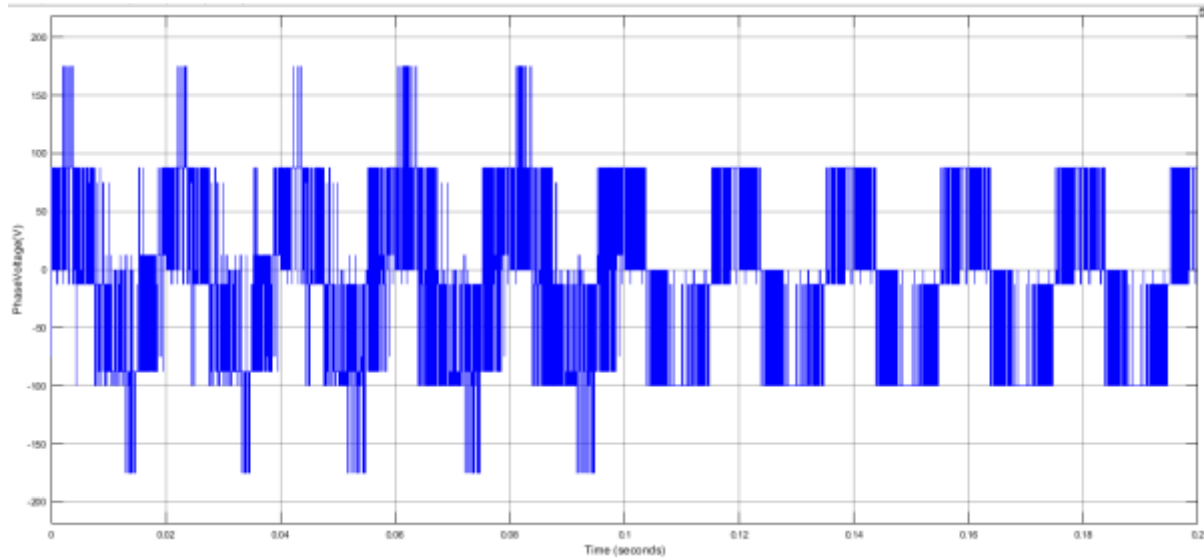


Primary H Bridge Capacitor Voltage

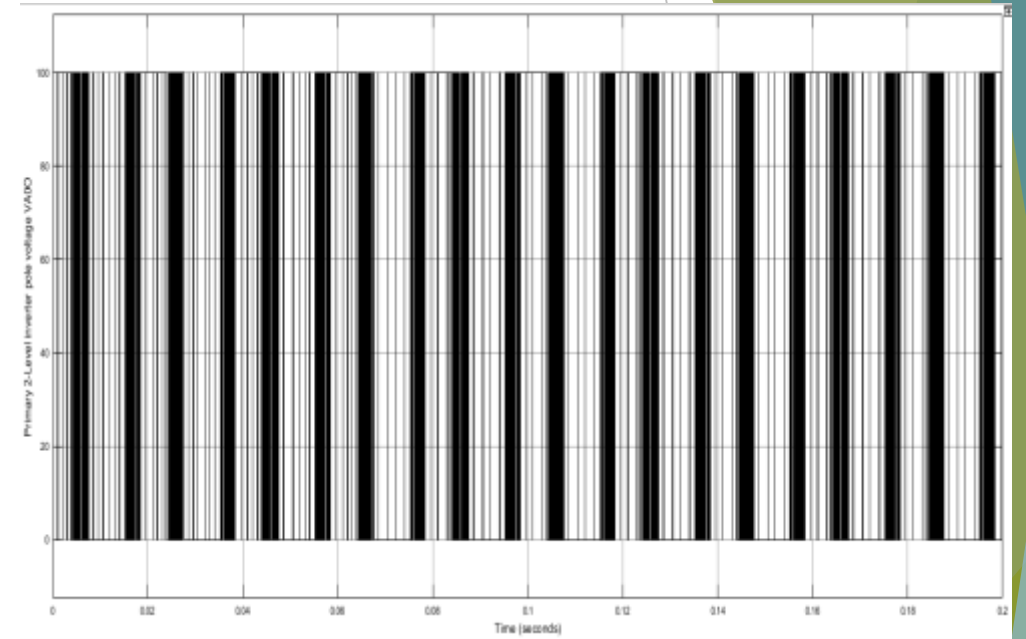


Phase current (Ia)

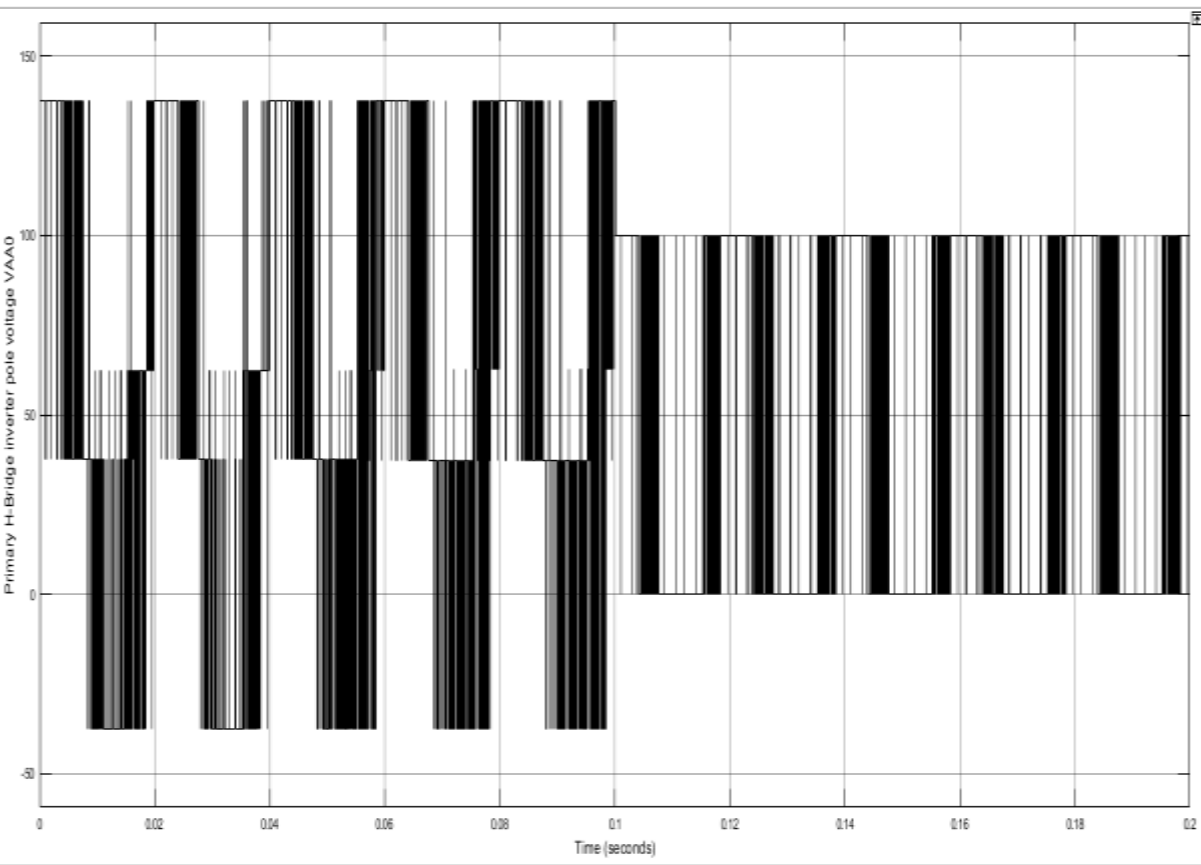
fault in 2-Level inverter at tf2 instant



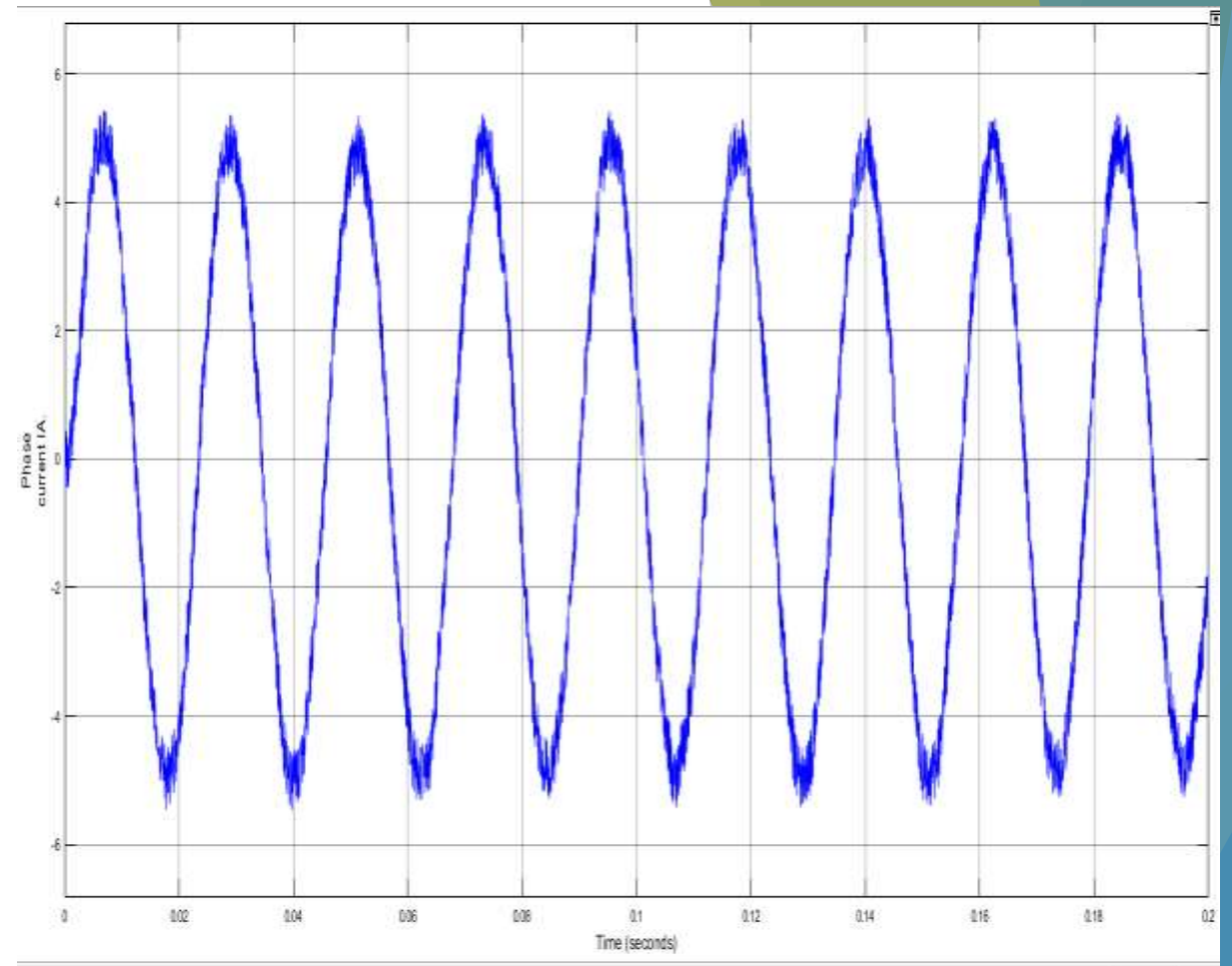
Phase voltage (VAA)



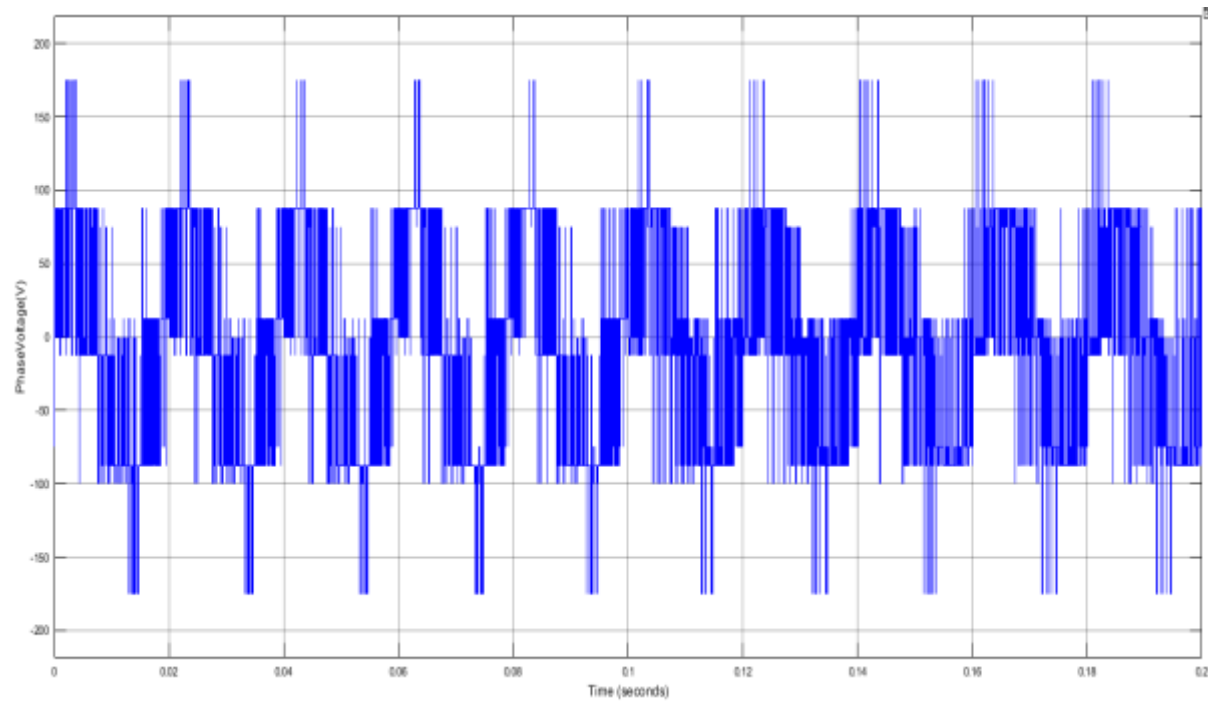
Primary 2-Level inverter pole voltage VA00



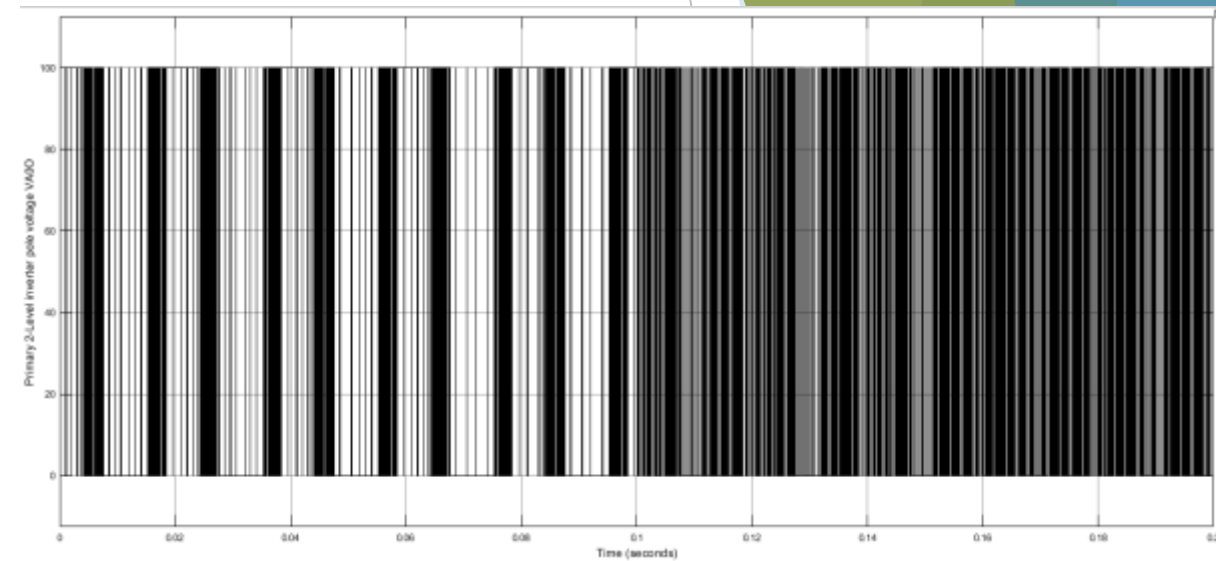
Hbridge inverter Pole voltage



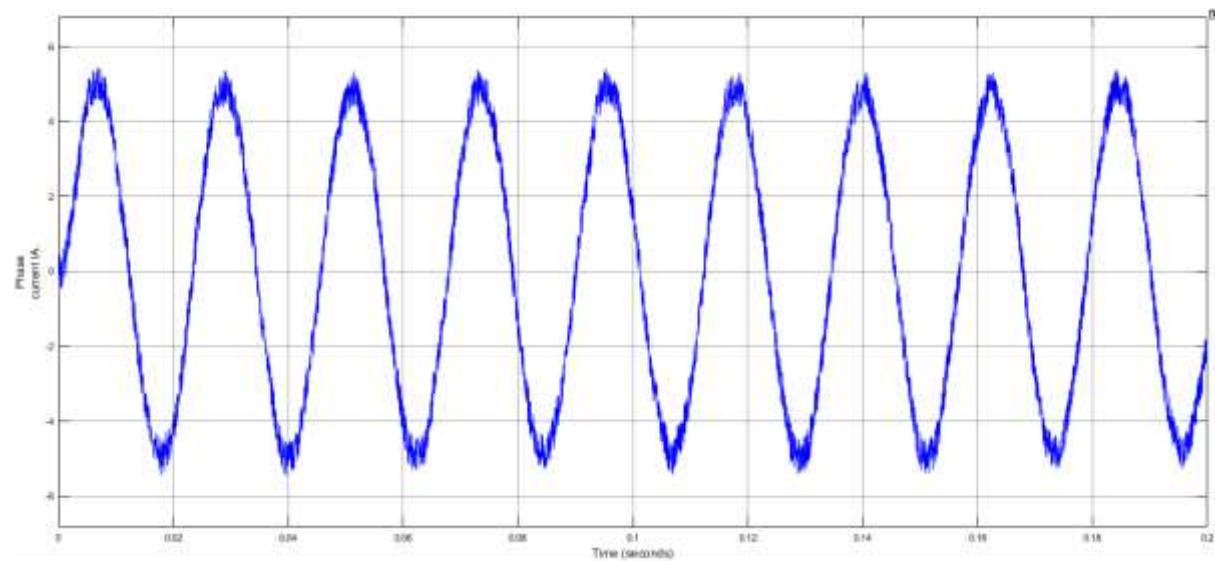
Phase current IA



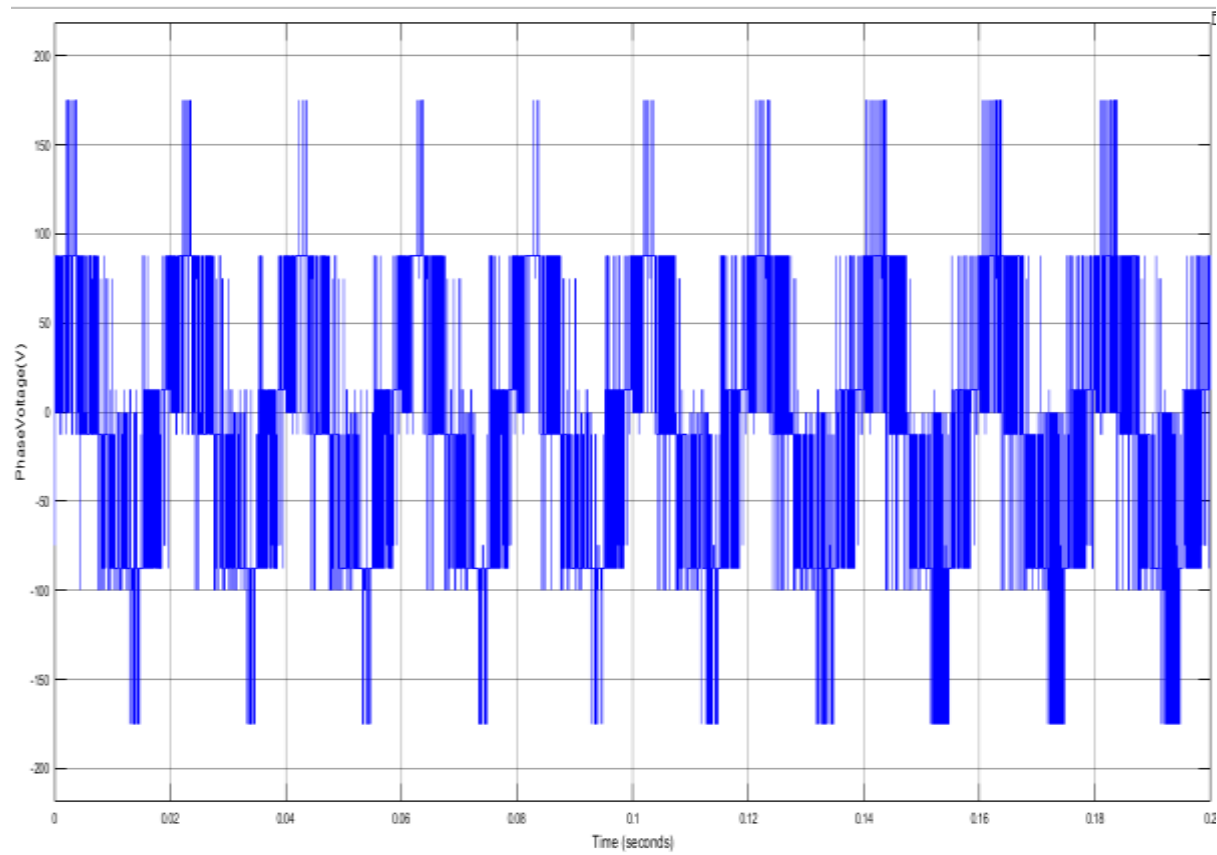
Phase voltage (VAA)



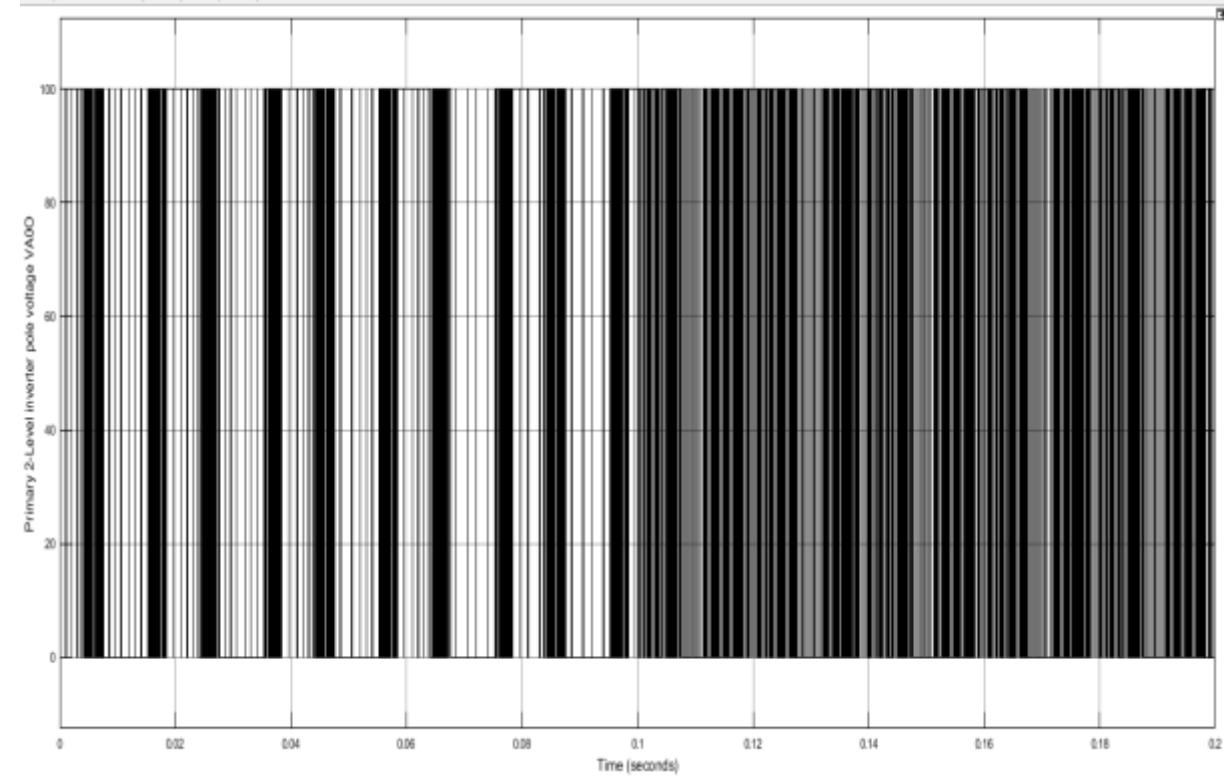
Primary 2-Level inverter pole voltageVA00



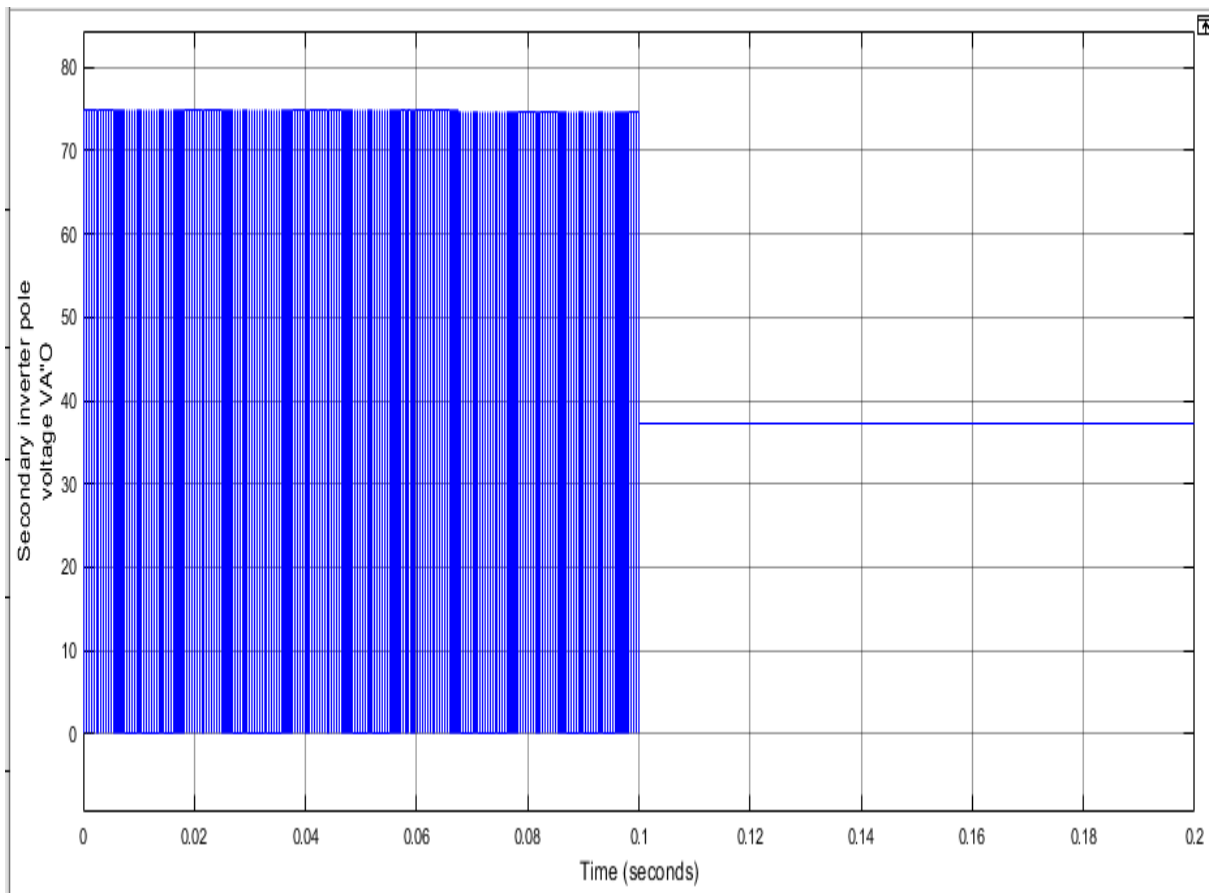
Phase current IA



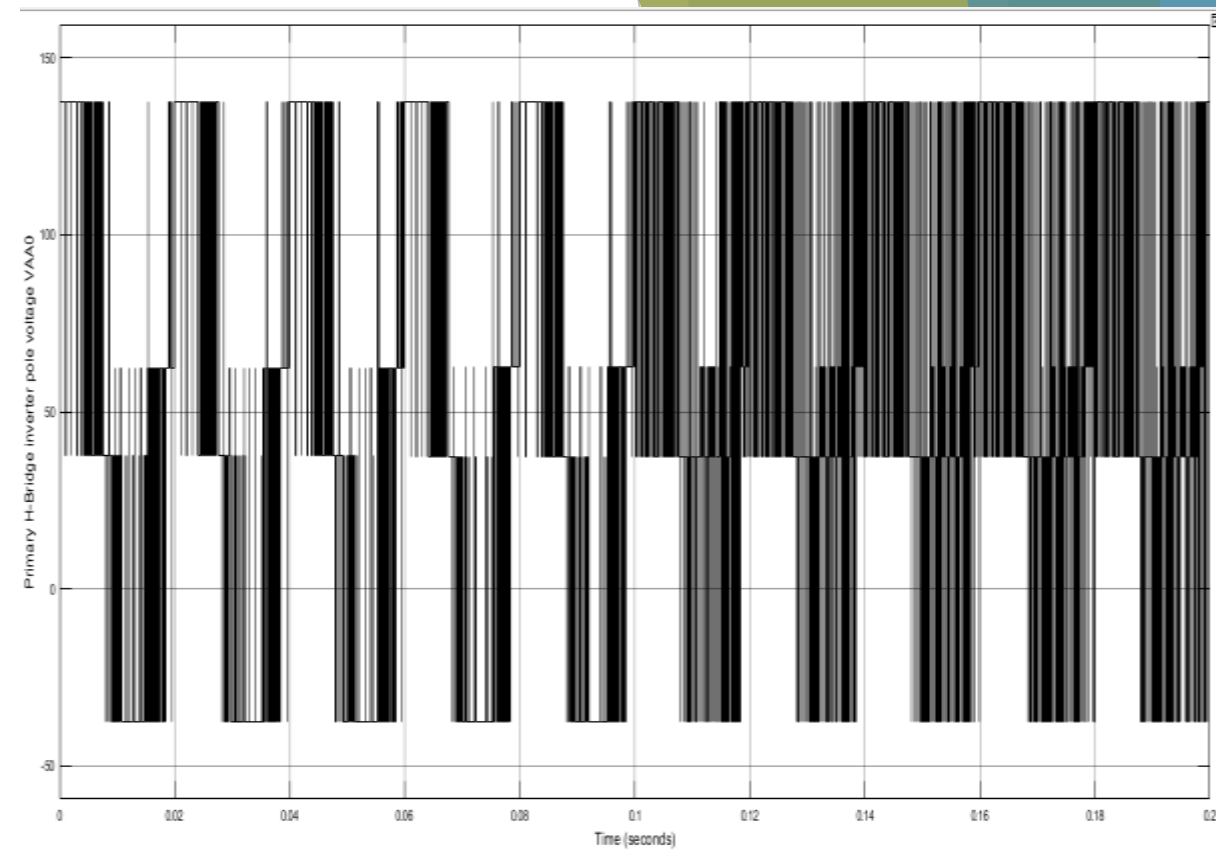
Phase Voltage



Primary 2 level voltage



Secondary inverter pole voltage



H bridge inverter pole voltage

CONCLUSION

- ❑ This paper proposes a reliable 5-Level inverter scheme with reduced component count for OEIM drive. Only one DC-link of voltage V_{dc} is required for the proposed scheme, which enables easy four quadrant operation. The inverter requires 4 capacitors, three are rated for $V_{dc}/4$ and one is rated for $V_{dc}/2$. The capacitor are balanced in each PWM cycle which allows to use low value capacitors in high power drive.
- ❑ The proposed scheme can run the drive even during fault in power switches in H-bridge and secondary 2-Level inverter. Comparison of the proposed inverter in terms of component count and switching loss with a few existing 5-Level inverter topologies are also discussed. As the single DC supply is connected to the primary 2-Level inverter, at the time of fault also drive can be operated at the rated power.
- ❑ Also, usage of single DC supply makes the scheme light weight and compact. Above mentioned features make the inverter compatible for high reliable industrial drives and electric vehicle applications.